

COMMON POISONINGS IN LIVESTOCK, THEIR TREATMENT & MATERIAL TO BE COLLECTED FOR TOXICOLOGICAL INVESTIGATIONS.

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Common poisoning occurs in animals and maybe divided roughly into:

I) poisoning by naturally-occurring toxicants and

II) Poisoning by man-made chemicals.

But there is no clear cut dividing line between these categories. Naturally occurring toxicological hazards include poisonous plants; man-made hazards include industrial contaminants, pesticides, domestic materials, unsuitable food and water, use of drugs, etc.

I) Naturally occurring toxicological hazards due to poisonous plants:-

Plants contain a large number of biologically active chemicals. Some of these have been found to be extremely useful for treating various human and animal diseases (e.g. digitoxin and atropine). However, some plant constituents produce adverse health effects following exposure. The onset of these adverse effects can be quite sudden or take some time to develop. Fortunately, among the thousands of plants in the environment of animals, relatively few cause acute, life-threatening illnesses when ingested. The magnitude of animal losses to ingestion of poisonous plants varies considerably between geographic regions. Losses include not only those due to mortality but also those due to poor productivity such as decreased weight gain or decreased milk or egg production. In addition, the economic cost of controlling poisonous plants needs to be considered.

Unfortunately, there are few antidotal therapies for treating plant poisonings. The best approach for treating intoxicated animals often involves routine decontamination procedures such as induction of emesis (in appropriate species) and the administration of activated charcoal and a cathartic to hasten elimination of the plant from the gastrointestinal tract. In addition, symptomatic and supportive care needs to be provided. Obviously, continued exposure to the suspect plant should be stopped. For a few plant poisonings, specific antidotes may be indicated; the treatment of cyanide or nitrate intoxicated animals are examples.

It has been estimated that in India there are about 700 poisonous species of plants belonging to over 90 families. Among these few plants are commonly encountered during grazing of livestock which can produce lethal toxicity to animals. In this I have included description of important poisonous plants, toxic principles, signs and symptoms, diagnosis and their treatment & material /samples to be collected for laboratory investigations.

1. Cyanogenetic Plants:-

Hydrocyanic acid is present in certain plants in the form of cyanogenic glycosides. Its consumption in large quantity within a short period of time can cause **Hydrocyanic acid poisoning**.

Cyanogenetic Plants: *Sorghum* sp. (Jowar) [Immature/ under developed], *Prunus* sp. (Apricot leaves), *Acacia leucophloea* (Hiwar/Shembati), *Eucalyptus* sp. (Nilgiri), *Nerium oleander* (Kaner), Sugarcane leaves, Shewanti, Palas, Takala, Dafal (Double bee) etc. Even crops/fodder heavily fertilized with nitrogen containing fertilizers promotes accumulation of cyanogenic glycosides in the crops/fodder & situation is further aggravated if rains are less.

Susceptible Animals: Cattle and buffaloes are more susceptible but less toxic to sheep and goats. During February, March pods of hiwar tree are shed largely on the ground. Hence farmers and sheep and goat rearers are advised not to graze their animals in areas adjacent to the Hiwar plantation. The carpels of Hiwar pod contains more quantity of HCN.

Symptoms: In acute case, following large quantities of ingestion of plant materials, animal may die within a few minutes. Death may be preceded by dyspnoea, staggering gait, difficulty in breathing, drooling of saliva, bright red mucosa and tremor.

Diagnosis: History of sudden death following grazing. Stomach contents may give odour of cyanide (bitter almond). Bright red blood and mucous membrane. Congested and haemorrhagic lungs. Detection of HCN in the suspected material and ruminal contents by picrate paper test.

Treatment: Prevent further exposure to cyanogenetic plants. Sodium nitrite and sodium thiosulphate are administered. Sodium nitrite (10 g/100 mL of distilled water or isotonic saline) should be given IV at 20 mg/kg body wt, followed by sodium thiosulfate (20%), IV, at 50 mg/kg; the latter may be repeated as needed with little hazard. Sodium nitrite therapy may be carefully repeated at 10 mg/kg, every 2-4 hr or as needed. Sodium thiosulfate alone is also an effective antidotal therapy at 50mg/kg, IV, plus 30 g/adult large animal, orally, to detoxify any remaining HCN in the rumen.

Prevention & control :- Pasture grasses should not be grazed until they are 15-18 inch tall to reduce danger from HCN poisoning. **Forage sorghums should be several feet tall. Never use Sorghum sp. (Jowar) for hydroponic.** Animals should be fed before first turning out to pasture; hungry animals may consume forage too rapidly, to detoxify HCN released in the rumen. Animals should be turned out to new pasture later in the day; HCN release potential is reported to be highest during early morning hours. Grazing should be monitored closely during periods of environmental stress, eg, drought or frost. Abundant regrowth of sorghum can be dangerous; these shoots should be frozen and wilted before grazing.

Material to be collected for laboratory confirmation:-

1. Ruminal, Intestinal content, Vital organs - liver, lung, muscle piece on Common salt & should be tightly packed & dispatched as soon as possible .
2. fodder, greens / plants suspected to be ingested by the animal.
3. **Vanderwalt (1944)** advocates the use of 1% solution of mercuric chloride & states that if liver tissue & rumen content are preserved in this way immediately removed from the body then loss of cyanide is almost completely prevented.

Sorghum vulgare



Acacia leucophloea / हिवर



हिवराच्या शेंगा

Images of plants with HCN Content

2. Botanical name: *Lantana camara*.

Common name: Lantana, wild sage, Bunch berry, Putus.

Hindi name: Raimuniya, Chaturang, Khaneri, Dhani-dadia.

Marathi name: - Ghaneri, Tantani.

Toxic Part: Leaves and green berries.

Toxic Principles: Lantadienes.

Susceptible animals: All farm animals, especially cattle, sheep and goats.

Symptoms: Ruminal stasis, constipation, icteric mucus membrane, swollen eyelids, photosensitization in light skinned animals.

Diagnosis: History of lantana feeding. Clinical symptoms like enlarged liver and distended gall bladder.

Treatment: Prevent further exposure to Lantana. Keep animals away from direct sunlight. Give activated charcoal to prevent further absorption. Hepatoprotective agents, antihistaminics and antibiotics can be given.



लॅन्टेना कैमेरा लक्षण:- अंगावरील कातडी तडकते व जखमा होतात

3.Nitrate and nitrite poisoning:

Nitrates are found in many areas of our environment. Almost all animals are susceptible to nitrate poisoning especially if they consume plants that have excess amounts of the compound. Plants absorb nitrates from the soil and metabolize them to form plant proteins. If plants absorb excess nitrates and are consumed by livestock before they are converted to proteins, nitrate poisoning can occur. Forage crops that are over fertilized before being harvested or grazed can be a common cause of nitrate poisoning. However, excess nitrate accumulation also occurs readily **in some common pasture weeds**. Nitrate concentration can vary widely among plants and growing conditions.

Nitrates are highest in plants in mornings and evenings, and **on cool, cloudy days (when plant metabolism is slower)**. **Drought, fertilization and nutrient deficiency** can result in nitrate accumulation in plant tissues. Highest concentrations occur generally in stems, rather than leaves, flowers or fruit/seed.

Cattle are the most commonly affected animals of nitrate poisoning. Runoff into ponds or other water supplies can contain high levels of nitrate thus potentially poisoning any animals that uses the source for drinking. Animals that ingest too much food containing nitrates or perhaps eat the actual fertilizer, can suffer from nitrate poisoning which is potentially deadly.

Sources: Nitrate containing plants (**turnip, linseeds, sugarbeet, common oat and maize**). Vegetables capable of accumulating large amounts of nitrate that are most frequently grazed include **Spinach, lettuce, cabbage, potatoes and carrots**. **Weeds** capable of nitrate accumulation include **pigweed Katemath (Amaranthus), Kena, lambsquarter and many others**. Other sources are fertilizers and feed, drinking of nitrate rich water.

Susceptible animals: Ruminants (cattle, buffaloes, sheep and goat) are commonly affected with nitrate poisoning. Non-ruminants (pig, horses and dog) are susceptible to nitrite but not to nitrate due to absence of rumen microflora.

Symptoms: Rapid respiration, salivation, colic, weakness, rapid and weak pulse, bluish mucus membrane, convulsion and death.

Diagnosis: History of consumption of nitrate/ nitrite containing materials. Clinical signs, **Dark brown to chocolate coloured blood and tissues**. Congestion of the intra-abdominal organs, petechial haemorrhages on the serous surface, **dilatation of the blood vessels**, generalised cyanosis and blood stained pericardial fluid are common **postmortem changes**. Detection of nitrite/nitrate in ruminal/stomach contents.

Treatments: Remove the sources of poison. Acute poisoning can be treated by slow intravenous injection of methylene blue at the rate of 4-8 mg/kg in cattle and sheep as a 1% solution or Ascorbic acid @ 15mg/kg IV.

Prevention & control: - Ruminant animals including cattle and sheep convert nitrate to nitrite which later turns into ammonia. When any crops have been exposed to fertilizer, they should be tested for the nitrate levels before using as a feed supply to livestock. Introduce suspected or high-nitrate forages gradually into the ration over a period of two to three weeks to allow for adaptation and reduce risks. Tanks that are used to haul fertilizer should not be used to transport any type of water supply. Owners of any type of animal should avoid using fertilizers where animals are known to graze or drink. Keep fertilizers stored away from pets and livestock to prevent accidental consumption. At the first signs of poisoning, owners should seek emergency care for their animal. Also feeding of crushed grains in larger proportion in gradual manner (feedstuffs that are high in energy may aid the conversion of nitrite to ammonia and reduce the risk of nitrate toxicity) and feeding vitamins and mineral supplements regularly helps to minimize risk of nitrate poisoning. Making silage from fodder containing higher levels of nitrates reduces nitrate levels upto

50%. But should be fed after nitrate testing and due caution. Desired Quantity should be attained gradually.

Material to be collected for laboratory confirmation:-

1. Ruminal, Stomach content, urine, Serum on ice (If you suspect Nitrate poisoning no need to send other tissues).
2. Suspected fodder, greens, plants suspected to be ingested by the animal should be packed in paper (not in plastic bags) for laboratory assay.
3. Suspected water sample in chilled condition.

The color of venous blood may be of great help for differential diagnosis between Cyanide & Nitrate/Nitrite toxicities as shown below:-



Differential colors of Blood

B – Cyanide intoxication
(Bright red)

A – Normal appearing
venous blood.

C – Nitrite intoxication
(Chocolate Brown)



3. Botanical name: *Ricinus communis*

Common name: Castor plant

Hindi name: Arandi

Marathi name :-Arand

Toxic Part: Seeds (castor bean) are more toxic.

Toxic Principles: Ricin

Susceptible animals: All farm animals especially horses.

Symptoms: Abdominal pain, bloody diarrhea, dehydration, muscular spasm, convulsion, incoordination and prostration.

Diagnosis: History of feeding castor beans, clinical signs and symptoms and haemorrhagic gastro-enteritis.

Prevention and Treatment: Prevent further exposure to castor plant. Give activated Charcoal to prevent further absorption and sodium bicarbonate to promote urinary excretion of ricin. Prevent dehydration. Anti-ricin serum is administered if available.



एरंड

शास्त्रीय नाव - रायसिन कम्युनिस
विषारी घटक - रायसिन

4. Botanical name: *Datura stramonium*

English name: Jimson weed, Thorn apple.

Hindi/ Marathi name: Dhatura.

Toxic part: Leaves, berries, seeds.

Toxic principle: Hyoscyamine, atropine, hyoscine.

Susceptible animals: Herbivores (horse, cattle, goat) are usually much more resistant than carnivores (dog and cat). Pigs are susceptible.

Symptoms: Dry mouth and mucous membrane, ruminal atony, constipation, abdominal pain, mydriasis, blurred vision, muscle trembling, ataxia, convulsion, respiratory failure can lead to death.

Diagnosis: History of feeding dhatura plants, presence of plant parts in rumen/stomach, clinical signs. A drop of urine from suspected case causes mydriasis when put in the eye of a cat. The mydriatic pupil does not constrict when exposed to light, while untreated eye constricts.

Treatment: Prevent further exposure. Gastric lavage may be given. No specific antidote is available. Usually symptomatic treatment is done. Physostigmine or pilocarpine is beneficial in recommended dose.



5. *Pteridium aquilinum*

Botanical name: *Pteridium aquilinum*

English name: Bracken fern.

Hindi/ Marathi name: Pharna.

Toxic part: Young plant leaves (fronds) and rhizomes.

Toxic principle: Ptaquiloside (also called aquiloside A) (ruminants), Thiaminase (nonruminants).

Susceptible animals: Both ruminants (cattle, sheep) and non-ruminants (horse, mule, pig).

Symptoms: Horse- staggering, tremor and serosal and mucosal haemorrhages. Cattle- dyspnoea, lack of rumination, high temperature, constipation, blood in faeces and dark urine (haemturia).

Diagnosis: History of access of bracken fern, clinical signs and symptoms.

Treatment: Withdraw the animals from contaminated area. Thiamine hydrochloride can be given daily up to two weeks in horse. In cattle, butyl alcohol (stimulates bone marrow), Tween 80, and sodium chloride can be given by slow intravenous injection. However, blood transfusion is the most effective remedy in cattle.



6. *Ipomoea carnea*

Botanical name: *Ipomoea carnea*

Common name: Bush morning glory

Hindi/ Marathi name: Besharam

Toxic part: Leaves are more toxic.

Toxic principle: Exact principle is not known. *Ipomoea carnea* leaves contain toxic saponins. *Ipomoea* sp. also contains resinous cathartic and hallucinogenic lysergic acid alkaloid. Ipomeanols (mycotoxins produced by fusarium fungi) can cause poisoning in sheep and goats.

Susceptible animals: All farm animals especially sheep and goat.

Symptoms: Salivation, diarrhea, mydriasis, staggering gait, ataxia, paralysis of limbs, anemia and death.

Diagnosis: History of grazing on ipomoea plants, clinical sign, presence of plant material in the stomach.

Treatment: Prevent further exposure to plants. Give activated charcoal, liver tonic and haematinics.



8.Botanical name: *Parthenium hysterophorus*

Common name: Congress grass, Parthenium

Hindi name: Gajar Ghas

Marathi name: Gajar gavat.

Toxic part: Leaves, stem and flowers

Toxic principle: Parthenin

Susceptible animals: All farm animals especially cattle, buffaloes, sheep and goat.

Symptoms: Usually diarrhea followed by skin lesions i.e., itching, erythematous eruptions, patchy depigmentation. Edema around eyelids and facial muscles. Ingestion of the weeds causes photosensitization with liver affections and skin reactions. It causes allergic dermatitis in humans.

Diagnosis: History of grazing in the weed infested area, clinical symptoms and presence of skin lesions.

Treatment: Remove animals from weed contaminated pasture. Shift to normal fodder. Antipruritic and antiseptic may give to treat skin lesions. Liver tonics may also be given.



9.Oxalis/Oxalic acid Poisoning:

Botanical name: *Oxalis sp.*

English name: Indian Sorrell

Hindi name: Tinapatiya, Amrulsak

Toxic part: Whole plant

Toxic principle: Oxalic acid

Susceptible animals: All farm animals, especially sheep.

Symptoms: Lowering of head, loss of appetite and isolation from herd are initial symptoms. Later frothy salivation, incoordination and coma with irregular respiration.

Following crops & vegetables also responsible to cause oxalate poisoning when fed in excess .

1.Animals grazing in the field of Onion, BT cotton, Tomato etc.

2.Before Vegetables taken to the market for sale, unwanted parts like plant leaves are fed to the animals, e.g. Cabbage, Cauliflower, beet, Spinach (Palak), Mula, Garlic, Ratale, Banana tree leaves, stem and roots, Pumpkin varieties, Sunflower beans, Lily flower plant leaves & tubers, Colocasia (Leaves of Alu) etc.

3.Importantly, feeding only and or maximum quantity of Sugar cane tops, Jowar (Sorghum), Maize plant, Maize shoot, without feeding the animals with sufficient quantity of dry fodder, highly increase the chances of oxalate poisoning.

Diagnosis:-

1.Oxalate poisoning has no characteristic postmortem signs but kidneys are usually swollen & pale & some animals will have severe fluid accumulations in the lungs with froth flowing from the trachea (windpipe).

2. GI irritation if eat large amount. Ex. experimental diets with more than 6% oxalates cause diarrhea in sheep - secondary fungal invasion of rumenitis and hepatitis lesions may follow.

3.Hypocalcaemia is characteristic – Removal of calcium from blood - precipitated as calcium oxalate, leads to a hypocalcemic muscle weakness syndrome with hemorrhages (Na oxalate is an anticoagulant because it binds calcium). Hypocalcemia contributes to great decrease in GI motility. Oxalates also have effect of immobilizing calcium in the gut if already on calcium deficient diet.This must be differentiated from parturient peresis, hypocalcaemia due to starvation or forced exercise and lactation tetany.

4. Acute nephrosis and uremia if calcium oxalate crystals precipitate in and block many tubules of the kidney - cumulative damage is possible; more tubules blocked with further exposure.

5.To help confirm oxalate poisoning in live animals, blood from the jugular vein can be analysed for its calcium content (very low in oxalate poisoning) and for indicators of kidney damage.

6.History of ingestion of known toxic plants.

7.Acute indigestion due to overeating of grains may cause similar syndrome. In such cases also history is important.

Treatment:-

1.Lime water orally @ 1 to 1.5 liters / large animal & for small animal @ 250ml. to 500ml / thrice in a day.

Lime water Preparation :- Add 1 kg lime in 10 Lit clean drinking water keep soaking for overnight. Remove supernatant without disturbing bottom lime layer. Use lime water judiciously.

2.In Animals suffering from a decrease in calcium and showing signs of hypocalcaemia may be treated with IV calcium.

Warning: - Overdosing with CBG can be fatal to stock. Inject slowly and monitor the pulse rate.

Dosage for a 25% solution of CBG

Cattle - 300-500 ml

Sheep- 50-100 ml

Calcium borogluconate:- IV or SC. Usually not enough to save animals because imbalance of other ions has occurred. Na^+ and K^+ liberated when Ca^{++} precipitated, may get alkalosis. Also oxalates have interfered with enzymes activated by calcium and magnesium, thus interfering with energy metabolism. This is the reason animals don't respond to calcium, but still try.

3. Ancillary: – Fluids to flush kidneys - dextrose solution IV, Provision of ample of fluids to decrease precipitation of oxalate crystals in urinary tract.

Control:-

1.Hungry animals should not be allowed to feed on large quantity of toxic plants.

2.If it is a history of the animals are grazing in the cotton field , it is advised that not to graze their animals in cotton fields.

3.Water intake affects the oxalate intake so give adlib water.

4.Alternative feed/fodder should be given, if the animal known to be ingesting oxalate containing feed/fodder or greens or if the symptoms of oxalate poisoning seen.

5.Prophylaxis – Give dicalcium phosphate, other calcium salts and bone meal are ineffective

6.Adequate salt should be made available.

7.Livestock should be adapted to oxalate plants over four days, incrementally increasing the time allowed to graze the plants, before being left in pastures containing high concentrations of oxalate-accumulating plants.

8.Therefore, if ruminants are slowly exposed to a diet high in oxalates (over an approximately 4 day period), the population of microbes degrading oxalate in the rumen increases sufficiently to prevent oxalate poisoning.

Samples to be submitted for laboratory confirmation –

1.Vital organs, Ruminal, Abomasal wall (on salt), Blood without anticoagulant, Serum (for detecting hypocalcemia), Urine (on ice) .

2.Fodder-green, semi-dry, dry or the plants suspected to be ingested by the animal. Samples of the plants should be collected for oxalate and nitrate assay. These should be packed in paper (not in plastic bags).



Indian Sorrel



तीनपानी

दलदल व पाणथळ जमिनीवर जोपासलेला चारा
यामध्ये ऑक्झलेट क्षाराचे प्रमाण अधिक असते,



काळसर पांढुरके ठिपके असलेला बुरशीयुक्त कडबा :- ऑक्झलिक ऑसिड

9. ONION POISONING :-

Toxic principle:

Organosulphide (N-propyl disulfide) compounds found in the [Allium family of vegetables](#) for example onion & garlic. Flesh, juice, leaves, raw, cooked, spoiled, and even processed powders, all contain the toxin. Both wild and cultivated onions contain the same toxic principle, N-propyl disulfide which primarily affects erythrocytes. The principle effects are related to hemolysis (rupture of red blood cells).

Mechanism of action:

Once in the bloodstream, the toxin enters red blood cells and causes oxidative damage to haemoglobin, rendering the red blood cell useless for carrying oxygen. Identified by the body and destroyed, the cells release their haemoglobin, initially into serum then filtered by the kidney and excreted in the urine, staining it pink in severe cases.

Clinical signs :

Urine discoloration may vary from red wine to almost black. Muscle weakness, rapid breathing, and a rapid heart rate may be noted if hemolysis is sufficiently severe. Some animals may be icteric (jaundiced) and/or have a characteristic onion odor to the breath. Loss of and appetite may also occur. Although severe toxicoses may be lethal.

Diagnosis:

Is based on symptoms of anaemia, falling red cell counts, haemoglobin in urine, or identification of damaged red cells under a microscope.

Treatment:

Activated charcoal can be given at 0.5 to 1 g/kg with a stomach tube . Whole-blood transfusions may be necessary for critically ill animals. Other supportive therapies depend on the animal's overall condition. Once the toxin is in circulation, antioxidant treatment using Vitamin C (20mg/kg, three times daily) can minimise the toxic damage to red blood cells.

Prognosis:

The vast majority of minor anaemias, associated with onion ingestion, go unrecognised by owner. Most animals can cope with minor falls in red cell count, typical of mild to moderate haemorrhage. If a large enough dose is absorbed, life-threatening anaemia may develop, falling to around 10% red cell count. Oxygen and blood transfusion may be required to sustain life until the bone marrow does its thing and replace the lost cells.

II) Poisoning by man-made chemicals.

Pesticide poisoning in animals

INTRODUCTION

A pesticide is usually defined as a chemical substance, biological agent, antimicrobial or disinfectant used against pests including insects, plant pathogens, weeds, mollusks, birds, mammals, fish, nematodes (roundworms) and microbes that compete with humans for food, destroy property, have a propensity for spreading or are a vector for disease or simply a nuisance. The term insecticide is used to denote agents assigned to kill only insects, but the term pesticide has a broader connotation and also includes herbicides, rodenticides, fumigants, nematocides, algacides, ascaricides, molluscicides, disinfectants, defoliants and fungicides.

HISTORY AND USAGE OF PESTICIDES

The first known pesticide was probably elemental sulphur dust used in Sumeria about 4500 years ago. In recorded history, nicotine sulphate was extracted from tobacco leaves for use as an insecticide in the seventeenth century. In the nineteenth century, pyrethrum derived from chrysanthemums (Shewanti), and rotenones derived from the roots of tropical vegetables were introduced. After its discovery in 1939 by Paul Muller, dichlorodiphenyltrichloroethane (DDT) found widespread use. However, with the recognition that it was a threat to biodiversity, its use has declined considerably.

In India, the use of pesticides began in 1948 with the introduction of DDT for the control of malaria and benzene hexachloride (BHC) for locusts. Production of these substances in India started in 1952.

The increase in pesticide use for agriculture has paralleled the increase in quality and quantity of food products over the years. At the same time, there has been an increase in the use of these products for deliberate self-harm (DSH). At times, pesticides have been accidentally consumed by animals and on rare occasions have even been used for malicious poisoning purposes.

1. Organophosphorus and Carbamate Pesticides poisoning.

Organophosphates Insecticides :-

Malathion, Sumithion, Diezinon, Neguvon, Azinphos-methyl, Dimethoate, Ethion, Phorate, Temephos, Thimathion etc.

Carbamate Insecticides:-

Aldicarb, benomyl, carbaryl, carbendazim, carbofuran, propoxur, triallate, Methiocarb, Mexacarbate etc.

SOURCE:-

Consumption of forage sprayed by Carbamate & Organophosphate Insecticide, contaminated drinking water & accidental intake.

MECHANISM OF ACTION:-

The toxicity of OP and carbamate pesticides is due to the disruption of the nervous system of an invertebrate or a vertebrate through the inhibition of cholinesterase (ChE) enzymes. These enzymes are involved in transmitting normal nerve impulses throughout the nervous system. An acute pesticide dose reduces the activity of ChEs, and nerve impulses cannot be transmitted normally. This can paralyze the nervous system, and it may lead to death, usually from respiratory failure.

EFFECT ON ANIMAL: -

Increased salivation, nausea, vomiting, colic, sweating, incoordination of urine & feces. Paralysis of skeletal muscles & dilatation of pupils. It is important to mention that all poisoned animals may not show all the clinical signs (as described above) with every OP or CM compounds. In other words, there is a great variation in symptomatology among the individual cases of poisoning. Surviving animals usually recover within 3–6 h with CMs and within 24 h with OPs.

Lesions: - There are no gross lesions on necropsy.

Treatment of poisoning: -

1. Before instituting antidotal therapy, monogastric animals, such as dog, should be given gastric lavage. Animals of any species can be given activated charcoal to stop further absorption of insecticides. Animals should be washed thoroughly with water if they are exposed to insecticides dermally. Intravenous (IV) fluid therapy is always beneficial. In the case of OP poisoning, antidotal treatment requires the combined use of atropine sulfate and pyridine-2- aldoxime methochloride (2-PAM). Atropine sulfate acts by blocking the muscarinic receptors from ACh. In ruminants, one-fourth of the total recommended dose (0.5 mg/kg) can be given as a slow IV injection, and the remainder through intramuscular (IM) or subcutaneous (SC) injection. The total dose of atropine sulfate for an average size horse is about 65 mg, and for a dog is about 2 mg. Atropine sulfate treatment can be repeated at an interval of every hour until all hyper-secretory signs have subsided. 2-PAM reactivates the AChE inhibited by OPs. The recommended therapeutic dose of 2-PAM is 20 mg/kg, IV. The injection of 2-PAM can be repeated once after 1 h at half of its initial dose. Care should be taken that only a freshly prepared solution of 2-PAM be used. It needs to be emphasized that the combined therapy of atropine sulfate and 2-PAM is superior to any other treatment till today in the case of OP poisoning. Although many other oximes have been tested against many OPs, none has been proven to be better than 2-PAM.

Unlike with OP poisoning, 2-PAM and other oximes are ineffective in CM poisoning cases. In fact, in the case of some CMs, such as carbaryl and carbofuran, 2-PAM therapy accentuates the toxicity. Therefore, atropine sulfate, with doses as described for OPs, is the only preferred antidote. When the animals are exposed to very higher doses of CMs, atropine sulfate does not appear to be a life saving antidote.



2. Chlorinated hydrocarbons poisoning :- DDT,BHC,Aldrin,Endrin,Endosulphate.

SOURCE:- Through intact skin.

MECHANISUM OF ACTION:-

- 1 Stimulation of CNS produce tremors, incoordinated walk, Convulsions, coma, respiratory failure and eventually death from respiratory paralysis.
2. Sensory nerve is most sensitive; nerve ganglia are least sensitive.
3. Produces a change in the permeability of the nerve membrane.
4. High body temperature.
5. Liver damage by high doses.

EFFECT ON ANIMALS: -

The chlorinated hydrocarbon insecticides are general CNS stimulants. They produce a great variety of signs-the most obvious are neuromuscular tremors and convulsions and there may be obvious behavioral changes common to other poisonings and CNS infections. Body temperature may be very high. Affected animals are generally first noted to be more alert or apprehensive. Muscle fasciculation occurs, becoming visible in the facial region and extending backward until the whole body is involved. Large doses of DDT, DOD, and methoxychlor cause progressive involvement leading to trembling or shivering, followed by convulsions and death.

Lesions: -

If death has occurred suddenly, there may be nothing more than cyanosis. More definite lesions occur as the duration of intoxication increases. Usually, there is congestion of various organs (particularly the lungs, liver, and kidneys) and a blanched appearance of all organs if the body temperature was high before death. The heart generally stops in systole, and there may be many hemorrhages of varying size on the epicardium. The appearance of the heart and lungs may suggest a peracute pneumonia and, if the animal was affected for more than a few hours, there may be pulmonary edema. The trachea and bronchi may contain a blood-tinged froth. In many cases, the CSF volume is excessive, and the brain and spinal cord frequently are congested and edematous.

Diagnosis: -

Chemical analysis of brain, liver, kidney, fat, and stomach or rumen contents is necessary to confirm the poisoning. The suspected source, if identified, should also be analyzed.

Treatment of poisoning: -

There are no known specific antidotes. When exposure is by spraying, dipping, or dusting, a thorough bathing without irritating the skin (no brushes), using detergents and copious quantities of cool water is recommended. If exposure is by ingestion, gastric lavage and saline purgatives are indicated. The use of digestible oils .such as com oil is contraindicated; however, heavy-grade mineral oil plus a purgative hastens the removal of the chemical from the intestine. Activated charcoal appears to be useful in preventing absorption from the GI tract. When signs are excitatory, a sedative anticonvulsant such as a barbiturate or diazepam is indicated. Anything in the environment that stresses the animal-noise, handling etc-should be reduced or removed if possible. If the animal shows marked depression, anorexia, and dehydration, therapy should be directed toward rehydration and nourishment either IV or by stomach tube. Residues in exposed animals may be reduced by giving a slurry of activated charcoal or providing charcoal in feed. Feeding phenobarbital, 5 g/day, may hasten residue removal.

Prevention & control: -

1. The farmers should be advised about the harmful effects of pesticides so that they should minimize the use of pesticides in crops. They should judiciously use the pesticide in terms of their quantity and frequency.
2. It should be assured that the pesticides produced in country should be distributed proportionately so that the indiscriminate and impropportionate use can be avoided.
3. All emphasis must be laid on the development of Bio-pesticides like viral, bacterial or fungal pesticides or pesticides of botanical origin like Neem or Tulsi or of cow urine based pesticides, which can be used in crops to kill the insect pests without polluting the environment
4. The harmful pesticides like some organochlorines, organophosphates and carbamates must be banned strictly in India, their production, import or use should be completely banned.
5. To reduce the effect of pesticide residues, some herbal preparations should be developed which can overcome the immunopathological, neuropathic or nephropathy effects, there are many herbs mentioned in our Indian ancient literature, which can be scientifically validated to prevent and control the harmful effects of pesticides. This will certainly give a new direction to the world not to depend on synthetic things.



3. Urea poisoning:-

SOURCE: -

Plants usually contain little or no urea. The most common source of poisoning of cattle and sheep is via ingestion of toxic levels in concentrate feed, due to an error in urea addition or due to uneven mixing. A less common source of poisoning is via direct ingestion from liquid preparations of urea, from bags of urea, or of urea granules concentrated on pasture due to spillage or uneven spread of fertilizer. Urea is also used as a means of ammoniation of hay, straw or whole-crop cereals. Excessive addition of urea or failure to convert to ammonia (NH₃) in the treated feed can lead to urea poisoning in animals.

MECHANISUM OF ACTION: -

In ruminants, nitrogen from urea is released in the rumen as ammonia and can be used by rumen microflora to synthesise protein. This protein then becomes available to the animal through the normal processes of digestion and absorption. However, if more urea is consumed than the rumen organisms can metabolise, the ammonia is absorbed from the rumen into the blood. The ammonia is then converted back to urea in the liver, and is then excreted by the kidneys. This pathway can easily be overwhelmed, when excess ammonia and urea circulate in the blood, causing poisoning. Poisoning can occur rapidly from a few minutes to four hours after consumption

EFFECT ON ANIMAL: -

The main signs of urea poisoning are due to the alkaline-corrosive effects of the high NH₃ levels in the digestive tract and other tissues and the neuro-toxic effect on the brain. Signs of poisoning arise within 10-30 minutes of ingestion of toxic doses. There are three main types of poisoning: acute (severe), subacute (moderate) and chronic (mild). Acute (severe) signs include salivation, severe abdominal colic, bloat, forced rapid breathing, muscle tremor, incoordination, bellowing, convulsions, violent struggling, weakness, recumbency, coma and death within 4 h of ingestion. Sub acute (moderate) signs include abdominal colic and milder nervous and circulatory signs. Chronic (mild) signs include poor appetite, listlessness and lethargy.

LESIONS:-

Animals decompose rapidly after death from urea poisoning and there are no specific signs of poisoning. Post-mortem examination immediately after death can show evidence of bloat, generalized congestion of the carcass, excess fluid in the pericardial sac, pulmonary oedema with excess stable white foam in the large airways and hemorrhages on the heart (epicardial and endocardial). There can be a marked ammonia smell when the rumen is opened. The pH of fresh rumen contents is a useful test that can be done in the field. An alkaline rumen (pH greater than 7.5-8) is suggestive of urea poisoning. histopathological signs, there appear to be inflammatory changes in the rumen, particularly in animals that may survive the initial poisoning but die or are euthanized a day or two later. Inclusion of formalin-fixed sections of rumen and reticulum from animals that die from suspected urea poisoning, will assist diagnosis.

DIAGNOSIS: -

The most useful diagnostic indicators are the history of access to urea and the signs shown by live, affected animals. Laboratory tests of blood samples are not very helpful, and no specific changes are seen at post-mortem examination. The following are general indicators of urea poisoning:

- History of access to urea.
- Laboratory testing of collected blood and rumen fluid immediately after death may indicate urea poisoning.

· Post-mortem – bloat; white foam in airways; ammonia odour when the rumen is opened; rumen pH 7.5-8.0.

Often a large pool of rumen fluid is seen on the ground at the nose of the beast. The animals usually suffer severe bloat and the fluid build up in gases forces the rumen fluid out through the mouth when the animal dies. Keep rumen and reticulum tissue samples in formalin for subsequent diagnosis.

TREATMENT OF POISONING: -

Treatment is rarely effective. However, if cattle can be handled, a stomach tube can be passed to relieve the bloat and then used to drench the animal with a large volume of cold water: 45 L for an adult cow is suggested, followed by 2-6 L of 5% acetic acid or vinegar. This dilutes rumen contents, reduces rumen temperature and increases rumen acidity, which all help to slow down the production of ammonia. Treatment may need to be repeated within 24 hours, as relapses can occur. Rumenotomy and removal of rumen contents is suggested for valuable animals.

PREVENTION & CONTROL: -

Urea should not be given to fasting or malnourished animals, or those with damaged livers. If urea is added to ruminant diets, it should be introduced gradually and increased to a safe level. Concentrate feeds with 3% urea or more in the DM should be regarded as high-urea feeds. Such feeds should be fed in restricted amounts, preferably spread over 12 hours or more of each day. Mixing urea with molasses, fodder/sugar beet or good quality silage reduces the risk of poisoning. Urea mixing in the feed should be thorough.



Collection, preservation and dispatch of material / samples to be sent to the laboratory for examination, in various poisoning -

Sr No	Cases with history of.....	Order of preference-Material to be sent	Preservative to be used	Quantity of sample required
1	2	3	4	5
1	Hydrocyanic acid (HCN) poisoning/ Cyanide poisoning	Ruminal, Intestinal content, Vital organs - liver, lung, muscle piece	Common salt	Vital organs - 250-500 gm, Fodder/green - 250-500 gm,
		Fodder, greens,	No preservative	Feed / fodder / green / concentrates / grains - 250-500 gm
		Blood	Potassium oxalate + Sodium fluoride	15-20 ml
2	Nitrate / Nitrite	Suspected water, fresh urine, feed, concentrate, mineral mixture fodder, greens, plants, serum, blood & vital organs, Bones, teeth	Immediately on ice within 1 - 2 hours OR on common salt/saturated salt solution	Vital organs - 250-500 gm, Feed / fodder/ green / concentrates / grains – 250-500 gm, Water - 200 ml, Urine - 15-20 ml, Serum - 5-10 ml,
3	Oxalate poisoning	Vital organs	Common salt / saturated salt solution	Fodder / plants / green - 250-500 gm, Urine - 15-20 ml, Serum - Vital organs - 250-500 gm,
		fodder, greens, plants	No preservative	Feed / fodder/ green / concentrates / grains – 250-500 gm.
		serum, fresh urine	On ice	Urine - 15-20 ml, Serum - 5-10 ml
4	Aflatoxicosis	Feed/fodder/concentrates/grains	No preservative	Feed / fodder/ concentrates / grains - 250-500 gm,
5	Zinc phosphide	Ruminal, Intestinal, stomach content, unopened crop, gizzard	In air tight container or sealed polythene bag	Ruminal, Intestinal, stomach content – 250-500 gm, and unopened crop, gizzard
6	Unknown poisoning	Ruminal /Intestinal/Stomach content, Vital organs - liver, kidney, lung, spleen,	Common salt	Vital organs - 250-500 gm, Feed / fodder / plants / green / concentrates / grains - 250-500 gm, Water - 200 ml
		suspected poisonous material, Feed, fodder, greens, concentrate, grains	No preservative	Feed/fodder/green/concentrates/grains/ suspected poisonous material - 250-500 gm,
7	Arsenic poisoning	Ruminal, Intestinal, stomach content	Common salt / saturated salt sol.	Ruminal, Intestinal, stomach content - 250-500 gm
8	Ammonia (Urea) poisoning	Ruminal, Intestinal, stomach content	On ice	Ruminal, Intestinal, stomach content - 250-500 gm, Suspected chemical - 50 ml, Feed - 250-500 gm, Fresh urine - 15-20 ml, Unclotted fresh blood - 15-20 ml
		Suspected chemical, feed, fresh urine, unclotted fresh blood (with preservative)	No preservative	Feed/fodder/concentrates/grains/ Suspected chemical-250-500 gm
9	Common salt poisoning	Ruminal / Intestinal / Stomach content, Vital organs - liver, kidney, lung, spleen, part of external ear	On ice	Ruminal /Intestinal/ Stomach content - 250-500 gm Vital organs - 250-500 gm, Feed - 250-500 gm, part of external ear
		Feed	No preservative	Feed / fodder / green / concentrates / grains - 250-500 gm,
10	Gaseous poisoning	Blood from ailing animal	Potassium oxalate + Sodium fluoride	Blood 15-20 ml